GRAVITATION

1. Kepler's law of areas is a consequence of :
(1) Conservation of linear momentum
(2) Net force on planet
(3) Conservation of angular momentum
(4) Constant angular speed

2. The total mechanical energy of Earth-Satellite system is :
(1) Positive
(2) Zero
(3) Negative
(4) Can have any value

3. Consider a satellite moving in a circular orbit around earth. Which of the following statements is wrong :
(1) It is a freely falling body
(2) It suffers no acceleration
(3) It moves with constant speed
(4) Its angular momentum is constant

4. Which of the following is incorrect statement:
(1) The motion of a particle under the central force is always confined to a plane
(2) For a central force, the position vector of the particle with respect to centre of force has a constant areal velocity
(3) Moon has no atmosphere because of high angular velocity
(4) Escape velocity on surface of moon is less than that of earth

5. Which of the following is correct for motion of an object under the gravitational influence of another object :
(1) Angular momentum is conserved
(2) Total mechanical energy is conserved
(3) Linear momentum is not conserved
(4) All of these

6. Escape speed of a body from the earth does not depend on :
(1) Mass of body
(2) Direction of projection
(3) Both (1) and (2)
(4) The distance of point of projection from centre of the earth.

7. A geostationary satellite :
(1) has a time period of 24 hours
(2) rotates from east to west
(3) can be in any plane
(4) can be at any height above surface of earth

8. If the orbital radius of satellite is decreased, then (PE = potential energy, KE = kinetic energy)
(1) PE decreases and KE increases
(2) Both PE and KE decreases
(3) Both PE and KE increases
(4) PE increases and KE decreases

9. In Kepler's third law, $T^2 = KR^3$, the constant K is :
(1) Higher for earth than mercury
(2) Same for all planets
(3) Higher for mercury than earth
(4) None

10. The dimensional formulae of gravitational potential and gravitational intensity are respectively :
(1) L^2T^{-2}, LT^{-2} (2) L^2T^{-2}, L^2T^2
(3) LT^{-2}, L^2T^{-2} (4) LT^{-2}, L^2T^2

11. Italian physicist Galileo recognized the fact that :
(1) bodies are accelerated towards earth with variable acceleration
(2) all bodies move towards earth with constant velocity when released from rest
(3) bodies with more mass have greater acceleration towards earth's surface
(4) all masses are accelerated towards earth with constant acceleration

12. Which of the following statements are true for "geocentric model" ?
(1) All celestial objects, stars, sun and the planets revolve around the earth
(2) Paths of all celestial objects were circular
(3) All planets and stars revolved around the sun
(4) Both 1 and 2

13. Geocentric model was proposed by :
(1) Galileo (2) Aryabhatta
(3) Ptolemy (4) Copernicus

14. A model in which the sun was the centre around which the planets revolved :
(1) is heliocentric model
(2) is geocentric model
(3) was mentioned by Aryabhatta
(4) Both 1 and 3

15. Who proposed a definite model in which planets moved in circles around a fixed central sun ?
(1) Nicolas Copernicus
(2) Aryabhatta
(3) Ptolemy
(4) Galileo

16. Kepler's laws were extracted from data compiled by :
(1) Tycho Brahe
(2) Newton
(3) Copernicus
(4) Aryabhatta

17. Sum of distances of a planet from two focii in an an elliptical orbit around sun is :
(1) constant
(2) equal to major axis of ellipse
(3) Equal to minor axis of ellipse
(4) Both 1 and 2

18. Kepler's law of area is valid for :
(1) Any conservative force
(2) Any central force
(3) Only gravitational force
(4) Only attractive force

19. A particle of mass m is placed inside a spherical shell of mass M at a point other than the centre :
(1) The particle will move towards the surface of shell
(2) The particle will move towards the centre
(3) The particle will remain stationary
(4) None of the above

20. Two artificial satellite of different masses are moving in the same orbit around the earth. Can they have same speed ?
(1) Yes
(2) No
(3) For heavy satellite speed is more
(4) None of these

21. Why does a tennis ball bounce higher at hills than on plains :
(1) Acceleration due to gravity is more at hill
(2) Acceleration due to gravity is same at hills and plains
(3) Acceleration due to gravity is less on hills
(4) None of these

22. An astronaut experiences weightlessness in a space satellite. It is because :
(1) The gravitational force is small at that location in space
(2) The gravitational force is large at that location in space
(3) The astronaut experiences no gravity
(4) The gravitational force is infinitely large at that location in space

23. When you go from equator to the poles the value of acceleration due to gravity :
(1) Increases
(2) Decreases
(3) Remains same
(4) Decreases upto a lattitude of 45° and then increases

24. Maximum value of gravitational potential energy is:
(1) At centre of earth (2) At surface of earth
(3) At infinity (4) None of these

25. Gravitational P.E. of body is negative, why ?
(1) Because of attractive force
(2) Because of repulsive force
(3) Because of friction force